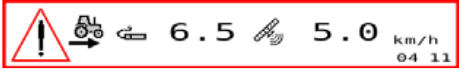
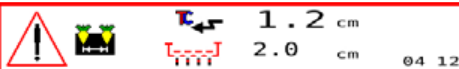
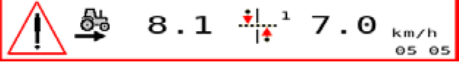
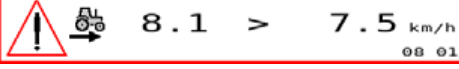
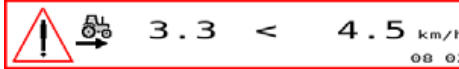


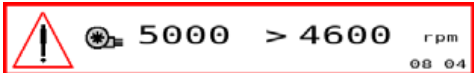
Optima E-Drive Software

Fault	Cause	Remedy
 STOP?? 	Limit switch triggered during seeding work. The machine is most likely working too low. Note: This symbol may also appear as a warning when raising the frame.	Check the depth setting of the machine, correct as necessary. Check the settings of the limit switch on the row.
  MEMORY DEFAULT  01 01	Alarm ID: 01-01 The entries in the memory could not be restored. The memory was reset to the standard values.	Re-enter machine data.
  MEMORY DEFAULT SLAVE  01 02	Alarm ID: 01-02 The entries in the slave computer could not be restored. The memory was reset to the standard values.	Re-enter machine data.
  ERROR : FPGA  01 03	Alarm ID: 01-03 Hardware fault on the main PCB.	Re-start Inform customer service
  ERROR : FPGA SLAVE  01 04	Alarm ID: 01-04 Hardware fault on the slave PCB.	Re-start. Check machine data. Inform customer service.
   !=   01 05	Alarm ID: 01-05 Different software versions in the main boxes.	Inform customer service.

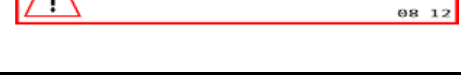
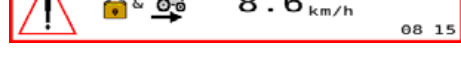
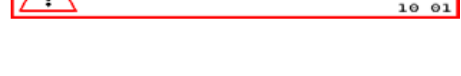
 ! IMPERIAL NOT SUPPORTED ! 01 08	Alarm ID: 01-08 Selected language or units of measurement of the terminal do not match the main box.	Select the supported language or measurement units on the terminal.
 2  02 01	Alarm ID: 02-01 Drive motor does not run. Seed disk is blocked.	Check connecting cable. Check fuses. Check seed disk and remove blockage.
 2  < 50 rpm 02 02	Alarm ID: 02-02 The seed disk turns too slowly or the driving speed does not match the sowing distance entered.	Check seed disk and remove blockage. Use a seed disk with fewer holes. Increase the driving speed. Reduce the sowing distance.
 2  > 215 rpm 02 03	Alarm ID: 02-03 The seed disk is turning too quickly or the driving speed does not match the sowing distance entered	Use a seed disk with more holes. Reduce the driving speed. Increase the sowing distance.
 2  rpm +/- > 15 % 02 04	Alarm ID: 02-04 Difference to the target speed.	Check seed disk and remove blockage. Check the drive motor. Inform customer service.
 2 I > 4 A 02 06	Alarm ID: 02-06 One of the motors is put under an average load of more than 4 A	Check the seeding heart for blockages Check the drive motor
 ECU - PWR 8.7 v 03 03	Alarm ID: 03-03 The supply voltage in the main box is too low	Check the battery Replace the fuse if the indicator shows "0"
  ECU - PWR 8.7 v 03 03	Short-circuit in individual sensors	Disconnect all sensors and reconnect them one after the other

  ECU - PWR 8.7 v 03 08	Alarm ID: 03-08 The supply voltage in the PEM main box is too low	Check the battery Replace the fuse if the indicator shows "0"
   1 : 00 h:m 04 01	Alarm ID: 04-01 TaskControl software test period is due to expire or has expired.	If you would like to continue to use the software, please purchase a license key.
    04 02	Alarm ID: 04-02 Changes to the machine settings with TaskControl selected.	Confirm the message to adopt changes in TaskControl. Choosing not to adopt the values can lead to undesirable seed results.
  04 05	Alarm ID: 04-05 It is not possible to use GEOCONTROL.	If you would like to continue to use the software, please purchase a license key. Deactivate GEOCONTROL.
   04 06	Alarm ID: 04-06 GEOSYNC receives no signals from the GPS antenna.	Check the antenna. Check the GPS settings Check the cables.
   04 07	Alarm ID: 04-07 GPS signal is too weak.	Check GPS antenna settings. Check GPS settings (GGA protocol).
  ≠ 10 Hz 04 08	Alarm ID: 04-08 The GPS signal refresh rate does not correspond to the 10 Hz required.	Set the GPS system to 10 Hz.
 ! NO AB LINE ! 04 09	Alarm ID: 04-09 No AB line has been created in the GEOSYNC software.	Create AB line.

	Alarm ID: 04-10 GEOSYNC receives no signals from the main PCB.	Check the cables. Inform customer service.
	Alarm ID: 04-11 The speed sensor detects a different speed to that calculated from the RTK signal.	Calibrate the speed sensor.
	Alarm ID: 04-12 The value for the application rate of TaskControl does not match the value of the application rate of the machine.	Check the application map. Check the basic settings of the machine. Inform customer service.
	Alarm ID: 05-04 The value entered for the application rate differs from the values for the GEOSEED calibration.	Enter the same value for both.
	Alarm ID: 05-05 The current speed differs from the speed entered for GEOSEED calibration by more than 1 km/h.	Maintain the speed via the tractor's automatic control system.
	Alarm ID: 08-01 The speed during sowing is higher than the set maximum speed.	Drive slower than the set maximum speed. Change the maximum speed input.
	Alarm ID: 08-02 The speed during sowing is below the set minimum speed.	Drive faster than the set minimum speed. Change the minimum speed input.

	Alarm ID: 08-03 No speed signal and the machine is in work position.	For the wheel sensor on the machine: Check that the wheel has permanent contact with the ground. Check the sensor connection cable. Test the sensor in the diagnostics menu.
	Alarm ID: 08-04 The fan speed is too high.	Keep the fan speed within the target range. For precision seeding machines, the fan speed is dependent on the vacuum required on the sowing units. The vacuum can be read on the manometer. → Operator's manual for the machine An entry for monitoring the fan speed is therefore only of limited use. → Chapter »e-drive II« software«, section »Basic settings«, page 24

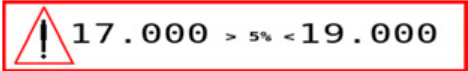
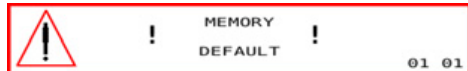
  3000 < 3400 rpm 08 05	Alarm ID: 08-05 The fan speed is too low.	Keep the fan speed within the target range. For precision seeding machines, the fan speed is dependent on the vacuum required on the sowing units. The vacuum can be read on the manometer. → Operator's manual for the machine An entry for monitoring the fan speed is therefore only of limited use. → Chapter »“e-drive II” software«, section »Basic settings«, page 24
  08 06	Alarm ID: 08-06 Fertilizer hopper almost empty.	Fill the fertilizer hopper.
   08 07	Alarm ID: 08-07 The metering device is not working.	Check the setting.
	The sensor on the metering device is defective.	Replace the sensor. Check the position of the magnets on the metering shaft.
	Blockage on the metering unit.	Check metering unit and remove blockage.

	Alarm ID: 08-09 Patchy seeding when depositing grain.	Check seed disk and remove blockage. Clean the opto-sensor. Replenish seed if necessary.
	Alarm ID: 08-10 Little seed in the seed hopper.	Replenish seed.
	Alarm ID: 08-11 There are no pulses on the sensor for synchronizing the sowing units.	Check seed disk and remove blockage. Inform customer service.
	Alarm ID: 08-12 Too many pulses for synchronizing the sowing units were detected.	Check sensor. Inform customer service.
	Alarm ID: 08-15 Driving with the machine locked.	Release the machine
	Alarm ID: 08-18 Doubles during grain deposit.	Check scraper settings. → Chapter »"e-drive II" software«, section »Scraper settings«, page 21
	Alarm ID: 08-19 Query whether the filling auger is already lifted.	Reject or confirm Chapter »"e-drive II" software«, section »Page 2«, page 14f.
	Alarm ID: 10-01 One or more sowing units do not work. Communication between master and slave disrupted.	Check the cables.

 SYNC CABLE MISSING 10 02	Alarm ID: 10-02 No signals coming from the synchronization cable.	Check the cables.
 MICRO VERSION MISMATCH 10 04	Alarm ID: 10-04 The software version of the microgranule distributor does not match the e-drive II software version.	Inform customer service.
 MICRO COMMUNICATION LOST 10 05	Alarm ID: 10-05 No communication with the software of the microgranule distributor.	Check the cables. Inform customer service.
  24 = 23 ? 11 01	Alarm ID: 11-01 Number of sowing units detected differs from the number of sowing units entered.	Enter the correct number. Check the sowing units.
  0 ! 11 02	Alarm ID: 11-02 Number of holes in the seed disk set at "0".	Enter the correct number.
  0 . 0 cm ! 11 03	Alarm ID: 11-03 Distance between the sowing units was specified incorrectly.	Enter the correct distance.

E-drive Fert software

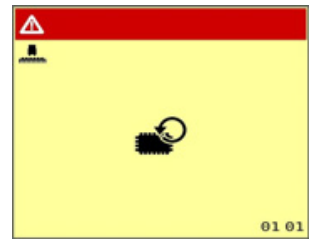
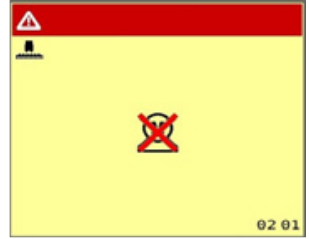
Fault	Cause	Remedy
 STOP?? 	No speed signal.	
	The machine is lifted.	For information only, not an alarm.
	Sensor defective.	Inform customer service.

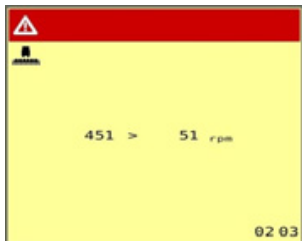
	The machine was lowered for longer than 30 seconds when stationary.	Release the drive motors.
	Reset to factory settings.	Confirm/Do not confirm
	Save these settings as factory settings.	Confirm/Do not confirm
	The difference in fertilizer applied between the right and left metering devices is greater than 5 %.	Check metering devices and drive motors.
	Alarm ID: 01-01 The entries in the memory could not be restored. The memory was reset to the standard values.	Re-enter machine data.
	Alarm ID: 02-01 Drive motor of the left/right metering device blocked.	Check the sensor setting Check that the sensor is working properly. Check the supply for the hydraulic drive motor.
	Alarm ID: 02-02 If the quantity of oil is too low, the number of revolutions on the left/right metering device is lower than the nominal value.	Increase the quantity of oil for the metering device. → Chapter »“e-drive II” software«, section »Basic settings«, page 73

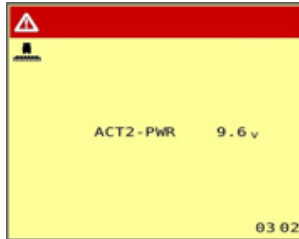
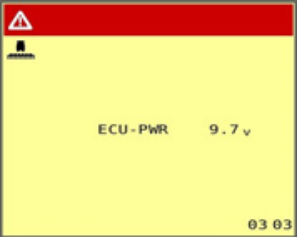
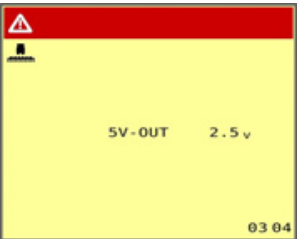
   31 < 82 rpm 02 02	Due to a blockage, the number of revolutions on the left/right metering device is lower than the nominal value.	Remove the blockage from the metering device.
   31 > 26 rpm 02 03	Alarm ID: 02-03 Due to an uneven load on the drive motors, the number of revolutions on the left/right metering device is higher than the nominal value.	Check the sensor setting. Check that the sensor is working properly.
 ACT - PWR 8.0 v 03 01	Alarm ID: 03-01 Voltage has fallen below the required voltage for the drive motors.	Check the battery. Replace the fuse if the indicator shows "0".
 ECU - PWR 8.0 v 03 03	Alarm ID: 03-03 Voltage has fallen below the required voltage for the main box.	Check the battery. Replace the fuse if the indicator shows "0".
 12V - OUT 8.0 v 03 05	Alarm ID: 03-05 Voltage has fallen below the required voltage for the 12 V consumers.	Check the battery. Replace the fuse if the indicator shows "0".
  04 02	Alarm ID: 04-02 Changes to the machine settings with TaskControl selected.	Confirm the message to adopt changes in TaskControl. Choosing not to adopt the values can lead to undesirable seed results.

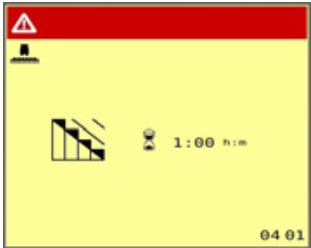
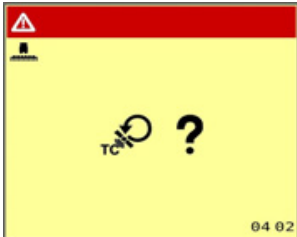
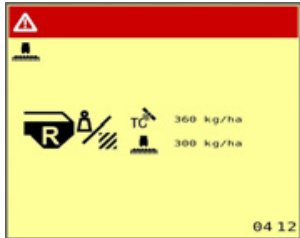
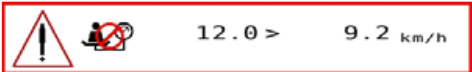
   80 kg/ha 200 kg/ha 04 12	Alarm ID: 04-12 The value for the application rate of TaskControl does not match the value of the application rate of the machine.	Check the application map. Check the basic settings of the machine. Inform customer service.
  05 01	Alarm ID: 05-01 Calibration test not correct.	Perform the calibration test again.
  4.0 m <  4.5 m 05 06	Alarm ID: 05-06 Current machine width and machine width from the selected calibration test do not match.	Carry out a new calibration test.
  25.0 > 20.0 km/h 08 01	Alarm ID: 08-01 The speed during sowing is higher than the set maximum speed.	Drive slower than the set maximum speed. Change the maximum speed input.
  1.5 < 2.1 km/h 08 02	Alarm ID: 08-02 The speed during sowing is below the set minimum speed.	Drive faster than the set minimum speed. Change the minimum speed input.
  L 5000 > 4600 rpm 08 04   R 5000 > 4600 rpm 08 04	Alarm ID: 08-04 Fan speed is too high.	Maintain fan speed in the target range.
  L 3000 < 3400 rpm 08 05   R 3000 < 3400 rpm 08 05	Alarm ID: 08-05 Fan speed is too low.	Maintain fan speed in the target range.

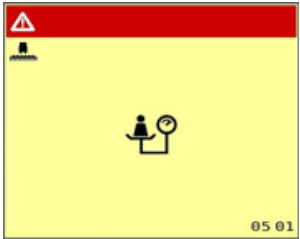
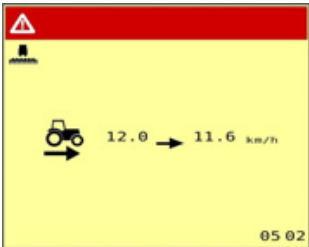
	Alarm ID: 08-06 Left/right fertilizer tank almost empty.	Fill left/right fertilizer tank.
	Alarm ID: 10-06 No connection to the seeding machine software.	Check the cables. Inform customer service.

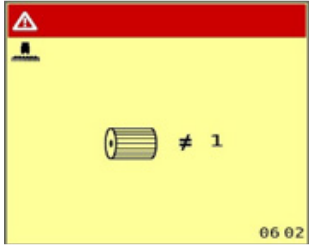
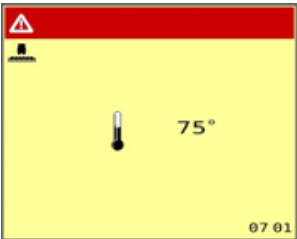
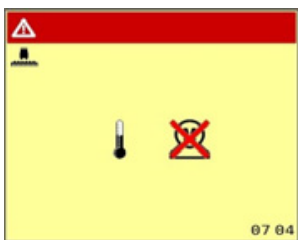
U-Drill Software		
Fault	Cause	Remedy
	Alarm ID: 01-01 All values have been reset to the factory setting Incorrect entries or errors on the board	Re-enter all inputs for the machine If the error occurs again, the board is defective and must be replaced.
	Alarm ID: 02-01 The metering device drive motor is rotating at a speed under 25 rpm	Check the metering device for foreign bodies

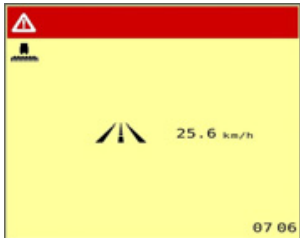
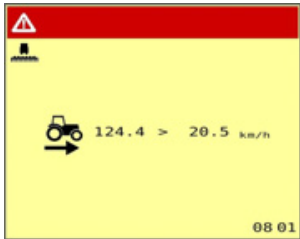
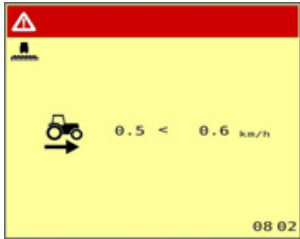
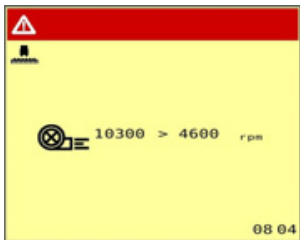
 Alarm screen 02 02: A yellow screen with a red header bar containing a warning triangle icon. The main display shows '48 < 448 rpm' and the alarm ID '02 02' in the bottom right corner.	Alarm ID: 02-02 The rotational speed of the metering device drive motor is falling below the set rotational speed by more than 10%	Check the metering device for foreign bodies
 Alarm screen 02 03: A yellow screen with a red header bar containing a warning triangle icon. The main display shows '451 > 51 rpm' and the alarm ID '02 03' in the bottom right corner.	Alarm ID: 02-03 The set rotational speed of the metering device drive motor is being exceeded by more than 10%	Check the metering device for foreign bodies
 Alarm screen 02 03: A yellow screen with a red header bar containing a warning triangle icon. The main display shows '451 > 51 rpm' and the alarm ID '02 03' in the bottom right corner.	Alarm ID: 02-05 Drive motor on the metering device is working too laboriously	Clean the metering device
 Alarm screen 02 06: A yellow screen with a red header bar containing a warning triangle icon. The main display shows 'I avr' and the alarm ID '02 06' in the bottom right corner.	Alarm ID: 02-06 Drive motor on the metering device is working too laboriously	Clean the metering device

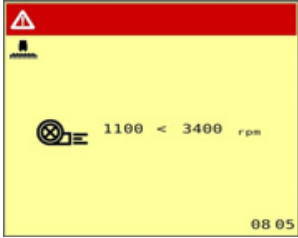
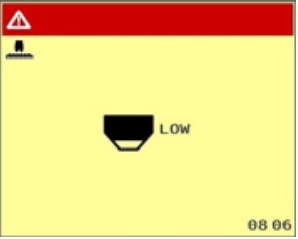
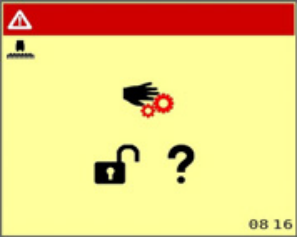
 The screen shows a red header with a warning triangle icon. Below it, the text 'ACT1 - PWR' is followed by '9.5 v'. At the bottom right, the code '03 01' is displayed.	Alarm ID: 03-01 Nominal current of actuator one is below 11 V	Check the cables! Check the cables for a short circuit if the indicator shows "0.0 V"
 The screen shows a red header with a warning triangle icon. Below it, the text 'ACT2 - PWR' is followed by '9.6 v'. At the bottom right, the code '03 02' is displayed.	Alarm ID: 03-02 Nominal current of actuator two is below 11 V	Check the cables! Check the cables for a short circuit if the indicator shows "0.0 V"
 The screen shows a red header with a warning triangle icon. Below it, the text 'ECU - PWR' is followed by '9.7 v'. At the bottom right, the code '03 03' is displayed.	Alarm ID: 03-03 Supply voltage required for the PCB (10 V) insufficient	Check the battery Check the cables of the connected sensors for a short-circuit if the indicator shows "0"
 The screen shows a red header with a warning triangle icon. Below it, the text '5V - OUT' is followed by '2.5 v'. At the bottom right, the code '03 04' is displayed.	Alarm ID: 03-04 Supply voltage required for the electrical system (4.5 V) insufficient	Check the battery Replace the fuse if the indicator shows "0"

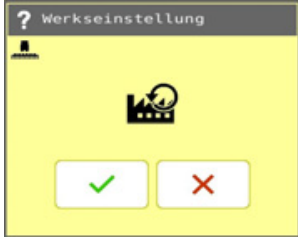
 The screen shows a yellow background with a red header bar containing a warning triangle icon. Below the header, there is a small icon of a person and a large icon of a person with a checkmark. To the right of these icons, the text '1:00 min' is displayed. At the bottom right corner, the text '04 01' is visible.	Alarm ID: 04-01 The time for the free TC section control testing is running out	If required, purchase a TC module and activate it If required, purchase a TC module and activate it
 The screen shows a yellow background with a red header bar containing a warning triangle icon. Below the header, there is a small icon of a person and a large icon of a person with a question mark. At the bottom right corner, the text '04 02' is visible.	Alarm ID: 04-02 Changes have been made to the settings; as a result, the TC module must be resynchronised	Confirm or discard the changes
 The screen shows a yellow background with a red header bar containing a warning triangle icon. Below the header, there is a small icon of a person and a large icon of a person with a checkmark. To the right of these icons, the text '360 kg/ha' and '300 kg/ha' is displayed. At the bottom right corner, the text '04 12' is visible.	Alarm ID: 04-12 The TC application page is above or below the completed calibration tests	Adjust the calibration test
 The screen shows a yellow background with a red header bar containing a warning triangle icon. Below the header, there is a small icon of a person and a large icon of a person with a checkmark. To the right of these icons, the text '12.0 >' and '9.2 km/h' is displayed.	Warning Calibration test: The required speed does not correspond to the inputs in the calibration test	Select the corresponding speed

	Alarm ID: 05-01 No active calibration test selected	Performing a calibration test
	Alarm ID: 05-02 Calibration test: The required speed is too high and will be adjusted	Confirm the alarm and travel at the specified speed
	Alarm ID: 06-01 Flap on the metering device is open during seeding	Close the flap
	Alarm ID: 06-01 Flap on the metering device is closed during calibration	Open the flap

	Alarm ID: 06-02 Wrong rotor installed	Replace the rotor
	Alarm ID: 06-03 The emptying function is activated	-
	Alarm ID: 07-01 Temperature on the PCB is higher than 70 °C	Switch the system off and then on again. If the temperature remains high, inform Customer Service.
	Alarm ID: 07-04 Temperature at the motor outlet too high	Confirm alarm. Then continue travelling.

 The image shows a yellow rectangular alarm display with a red header bar. In the top left corner, there is a warning triangle icon and a small tractor icon. In the center, there is a speedometer icon with the text '25.6 km/h' next to it. In the bottom right corner, the text '07 06' is displayed.	Alarm ID: 07-06 Driving speed > 25 km/h. All hydraulic functions are deactivated.	Reduce the driving speed
 The image shows a yellow rectangular alarm display with a red header bar. In the top left corner, there is a warning triangle icon and a small tractor icon. In the center, there is a tractor icon with a right-pointing arrow, followed by the text '124.4 > 20.5 km/h'. In the bottom right corner, the text '08 01' is displayed.	Alarm ID: 08-01 The maximum driving speed from the calibration is being exceeded	Reduce the driving speed
 The image shows a yellow rectangular alarm display with a red header bar. In the top left corner, there is a warning triangle icon and a small tractor icon. In the center, there is a tractor icon with a right-pointing arrow, followed by the text '0.5 < 0.6 km/h'. In the bottom right corner, the text '08 02' is displayed.	Alarm ID: 08-02 The driving speed is falling below the minimum driving speed from the calibration	Increase the driving speed
 The image shows a yellow rectangular alarm display with a red header bar. In the top left corner, there is a warning triangle icon and a small tractor icon. In the center, there is a fan icon, followed by the text '10300 > 4600 rpm'. In the bottom right corner, the text '08 04' is displayed.	Alarm ID: 08-04 The set fan speed is being exceeded by more than 10%	Reduce the fan speed

	Alarm ID: 08-05 The fan speed is falling below the set fan speed by more than 10%	Increase the fan speed
	Alarm ID: 08-06 Seed hopper is almost empty	Fill the seed hopper
	Alarm ID: 08-15 Machine released and speed signal shown, but still locked	-
	Alarm ID: 08-16 Rotating machine components	Stay clear of the working area of the machine

	Note: Using the actuator diagnosis can damage the machine Danger of injury due to crushing.	Confirm/Do not confirm
	Note: Using the actuator diagnosis can damage the machine Danger of injury due to crushing.	Confirm/Do not confirm
	Note: Reset to factory settings	Confirm/Do not confirm
	Note: Save these settings as factory settings	Confirm/Do not confirm